

UniConvNet: Expanding Effective Receptive Field while Maintaining Asymptotically Gaussian Distribution for ConvNets of Any Scale

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Abstract

Convolutional neural networks (ConvNets) with large effective receptive field (ERF), still in their early stages, have demonstrated promising effectiveness while constrained by high parameters and FLOPs costs and disrupted asymptotically Gaussian distribution (AGD) of ERF. This paper proposes an alternative paradigm: rather than merely employing extremely large ERF, it is more effective and efficient to expand the ERF while maintaining AGD of ERF by proper combination of smaller kernels, such as 7×7 , 9×9 , 11×11 . This paper introduces a Three-layer Receptive Field Aggregator and designs a Layer Operator as the fundamental operator from the perspective of receptive field. The ERF can be expanded to the level of existing large-kernel ConvNets through the stack of proposed modules while maintaining AGD of ERF. Using these designs, we propose a universal model for ConvNet of any scale, termed UniConvNet. Extensive experiments on ImageNet-1K, COCO2017, and ADE20K demonstrate that UniConvNet outperforms state-of-the-art CNNs and ViTs across various vision recognition tasks for both lightweight and large-scale models with comparable throughput. Surprisingly, UniConvNet-T achieves 84.2% ImageNet top-1 accuracy with 30M parameters and 5.1G FLOPs. UniConvNet-XL also shows competitive scalability to big data and large models, acquiring 88.4% top-1 accuracy on ImageNet. Code and models are publicly available at <https://github.com/ai-paperwithcode/UniConvNet>.

1. Introduction

With the impressive triumph of transformers [19, 38], constructing long-range dependencies has become a crucial principle in designing convolutional neural networks (ConvNets). Some prior works [17, 18, 37] have made attempts to capture relationships across large receptive fields, surpassing traditional convolutional neural networks [23, 45, 49, 59, 82] and achieving significant improvements in var-

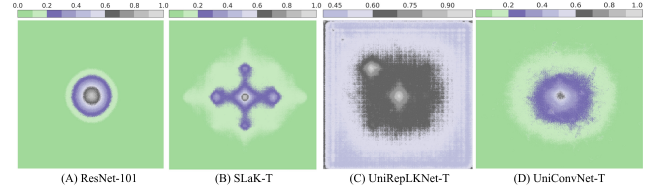


Figure 1. Effective Receptive Field (ERF) of ResNet-101, SLaK-T, UniRepLKNNet-T, UniConvNet-T. The more stepped colour area from the center indicates better asymptotically Gaussian distribution (AGD) of ERF. The wider area indicates a larger ERF. Large-kernel ConvNets, such as SLaK-T and UniRepLKNNet-T, disrupt the AGD of ERF.

ious vision recognition tasks, such as image classification, object detection, instance segmentation, and semantic segmentation. Current ConvNets achieve long-range dependencies by scaling up the convolutional kernel with re-parameterization [15, 17], parameter sharing [5] or sparsity [37] techniques. Some recent works leverage the key properties of large kernels [18] or encode their interactions [32] to inform ConvNet architecture design. They benefit from large ERF while constrained by high parameters and FLOPs costs.

A typical paradigm [21, 23, 70] for ConvNets is to use a stack of many small spatial convolutions (e.g. 3×3) to enlarge the receptive fields in ConvNets. Why small-kernel ConvNets constrained by small ERF still get effective performance? The conventional ConvNets, such as ResNet-101 [23], have a small ERF but benefit from multi-scale impact (gradient), which follows AGD, through the stack of 3×3 convolution modules, as shown in Fig. 1 (A). This suggests that smaller-scale pixels, around the position of the output pixel, of the input should have more impact on the output pixel. Large-kernel ConvNets, such as SLaK-T [37] and UniRepLKNNet-T [18], achieve a large ERF but disrupt the AGD, which either obtain a discriminative impact at weird position or get similar impacts of different scales, as depicted in Fig. 1 (B) and Fig. 1 (C).

Is there a proper way to combine smaller kernels to ex-

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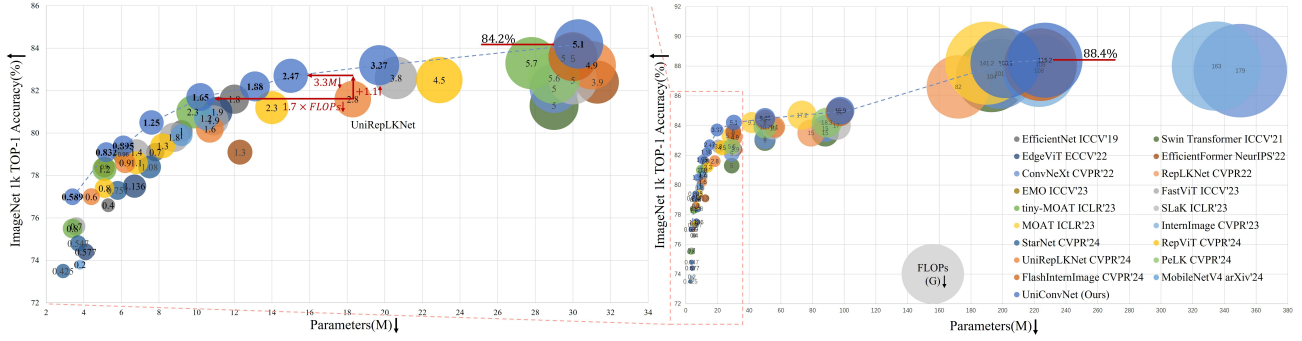


Figure 2. **Comparison of parameters and accuracy between UniConvNet (ours) and others.** The area of the circle, the number in the circle, represents the FLOPs of the relevant model. UniConvNet achieves the best accuracy-parameter and accuracy-FLOPs trade-off.

pand ERF while maintaining the AGD of ERF? This paper proposes an alternative paradigm: rather than merely employing extremely large ERF, it is more effective and efficient to expand the ERF while maintaining the AGD of ERF by proper combination of smaller kernels.

To answer this question, we introduce a Receptive Field Aggregator (RFA) for ConvNets, designed to obtain AGD at a shadow module by directly assign impact for different scales. The input images are separated into multiple heads according to the layer of RFA. Parameters and FLOPs costs are reduced by recursively feeding multi-head inputs into layer operators, creating a pyramidal increment among channels. For heads with different patterns among channels, in each layer, we propose a spatial encoder from the perspective of receptive field, called the Layer Operator (LO). The LO consists of two components: the Amplifier (Amp) and the Discriminator (Dis). The Amp expands the scale of the receptive field and amplifies the impacts of pixels on the receptive field by element-wise multiplication. The salient pixels in the receptive field will have a more distinguished impact. The Dis provides small-scale impacts from new pixels to the large receptive field produced by the Amp. The final receptive field becomes a large two-layer receptive field. Sequentially, each LO expands and amplifies the receptive field of the previous LO by Amp and provides a discriminative receptive field for adding impacts of small-scale pixels. The final receptive field of three-layer RFA results in a four-layer receptive field following an AGD. The ERF can be expanded and the AGD of ERF can be maintained through the stack of many RFA modules, as illustrated in Fig. 1 (D).

With these designs, the proposed UniConvNet efficiently reduces parameters and FLOPs while getting a multi-scale impact on ERF. Consequently, it outperforms state-of-the-art CNNs and ViTs in various vision recognition tasks, from lightweight to large-scale models, as illustrated in Fig. 2.

Notably, UniConvNet-T achieves 84.2% TOP-1 accuracy, surpassing models with similar parameters and FLOPs by at least 0.6 points, representing a significant improvement over existing ConvNets [5, 37, 40, 72]. UniConvNet-XL breaks through ConvNet bottleneck, achieving 88.4% TOP-1 accuracy with a superior parameters and FLOPs trade-off compared to contemporary CNNs [17, 32, 40, 50, 69, 72] and ViTs [13, 38, 39, 74]. UniConvNet is also powerful on downstream tasks. UniConvNet-L obtains 55.7% on COCO [34] and 55.1% on ADE20K [85].

We believe the high performance of UniConvNet is mainly because of the large ERF[42] while maintaining AGD as compared in Fig. 1. The ERF scale is comparable with ConvNets using extremely large kernels. The AGD of ERF are more similar to the intuition that the closer pixels should have more impacts. We hope our findings can help to understand the intrinsic mechanism of ConvNets.

2. Model Architecture

2.1. Receptive Field Aggregator

In order to expand the effective receptive field (ERF) [42] while maintaining the asymptotically Gaussian distribution (AGD) of ERF, we introduce a Receptive Field Aggregator (RFA) as illustrated in Fig. 3 (left), which directly assign discriminative impact on receptive field of different scales at a shadow module. Specifically, input images are initially divided into $N + 1$ parts along the channel dimension based on the layer N of RFA resulting in $N + 1$ heads: $A_1, H_1, \dots, H_n, \dots, H_N$. The input heads can be categorized into two parts: A_1 and $H_1, \dots, H_n, \dots, H_N$. Concretely, the input head $A_1 \in \mathbb{R}^{B \times \frac{C}{N+1} \times H \times W}$, where $B, \frac{C}{N+1}, H$, and W represent the batch size, channel dimension, height, and width, respectively, is initially fed into the Layer Operator (LO) 1, resulting in a new head A_2 . The channel dimension of the output head A_2 increases from $\frac{C}{N+1}$ to $\frac{2}{N+1}C$. Sub-

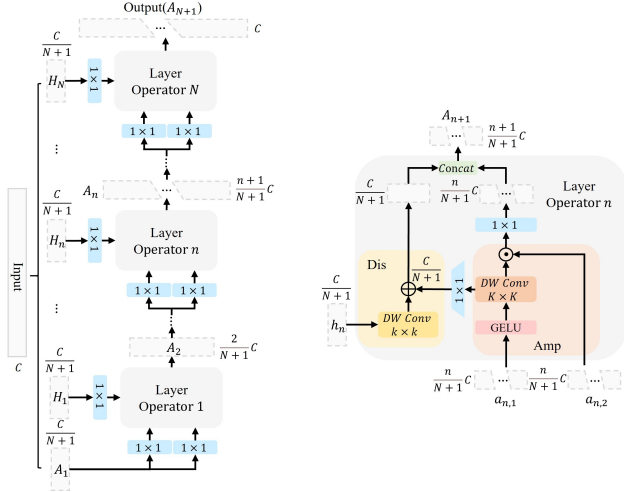


Figure 3. **Left:** Receptive Field Aggregator. **Right:** Layer Operator.

sequently, the output head A_2 is recursively fed into the LO n , illustrated in Fig. 3 (right), according to the layer number n ($n \in [2, N]$), with its channel dimension increasing from $\frac{n}{N+1}C$ to $\frac{n+1}{N+1}C$. The remaining N input heads, $H_1, \dots, H_n, \dots, H_N$ are sequentially fed into the LO n according to the layer number n ($n \in [1, N]$) to interact with the corresponding input head A_n . Each head is initially projected using 1×1 convolution before feeding in the LO to enhance feature diversity. In the RFA, the output channels of heads A_n follow a pyramidal increment, reducing parameters and FLOPs compared to the standard direct-in, direct-out construction. Increasing the layer N allows for higher input image resolution, potentially providing a more effective alternative to training on low-resolution images followed by fine-tuning at high resolutions.

2.2. Layer Operator

To effectively expand receptive field and assign discriminative impact on receptive field, we introduce the Layer Operator (LO). This operator is designed from the perspective of receptive field and serves as the core operator in the RFA, illustrated in Fig. 3 (right). This technique can construct a two-layer AGD of receptive field. Specifically, for layer number n , the three distinct inputs for LO n are $a_{n,1}$, $a_{n,2}$, and h_n , projected by three individual 1×1 convolutions as illustrated in Fig. 3 (left). The LO is generated by interacting two components, the Discriminator (Dis) and the Amplifier (Amp), as illustrated in Fig. 3 (right). For the Amp, we conduct an element-wise multiplication between $a_{n,2}$ and output features derived from a GELU activation and a depth-wise large-kernel $K \times K$ convolution applied to $a_{n,1}$. For each output pixel in the Amp module, the corresponding pixels within the relevant $K \times K$ receptive field

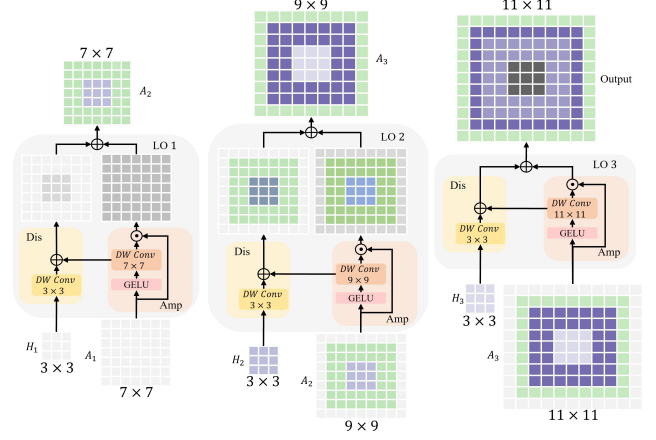


Figure 4. **Receptive Field Flow of the Three-layer Receptive Field Aggregator.**

are multiplied by the pixel value at the same spatial location in $a_{n,2}$. This operation expands the receptive field and amplifies the impact of pixels on the receptive field. Additionally, for the Dis, we incorporate features from depth-wise $K \times K$ and $k \times k$ convolutions. This introduces impact from small-scale new pixels for the large $K \times K$ receptive field, establishing a two-layer discriminative AGD. The 1×1 convolutions facilitate information interaction among channels and change the channel dimension for the compatibility of features. The outputs of the Amp and the Dis are concatenated, resulting in the final output A_{n+1} with a two-layer AGD of receptive field and increased channels for the subsequent layer.

2.3. Three-layer RFA for UniConvNet

The layer number N varies with the resolution of the input image. In this work, for the input image size of 224×224 , we construct a three-layer RFA using a layer $N = 3$. The progressive large-scale kernel size K in the LO is calculated as $K = 2n + 5$ ($n \in [1, N]$). The small-scale kernel size is $k = 3$. Thus, the RFA achieves a four-layer AGD for receptive field within a shadow module using progressive large-kernel convolutions 7×7 , 9×9 and 11×11 , which is an optimal configuration for 224×224 images. We do an ablation study in Sec. 4.1.

The smallest kernel size of 7×7 provides a significantly larger receptive field than 3×3 and 5×5 convolutions. The largest convolution kernel size of 11×11 can maintain the 14×14 features with padding size 5 in stage 3, the main stage of the feature extraction in our models, depicted in Fig. 5, of our model to have at most quarter pixels of the feature at the corner for avoiding the frequent overlapping of the center pixels during the convolution progress. The efficiency and effectiveness are justified and analyzed in Sec. 4.2.

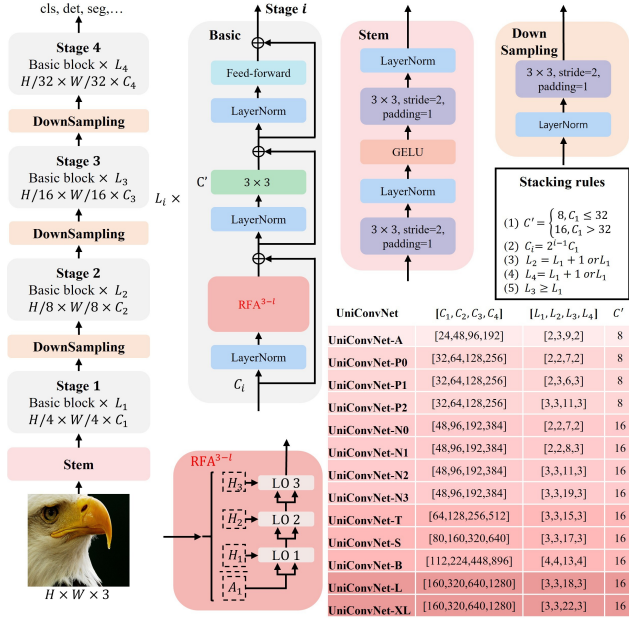


Figure 5. Overall architecture of UniConvNet.

2.4. Receptive Field Flow of Three-layer RFA

We sketch the receptive field flow of the proposed three-layer RFA in Fig. 4. In LO 1, the receptive field scale is expanded and the impacts of pixels on 7×7 receptive field is amplified by the Amp module. The Dis module provide the impacts of small-scale new pixels, closer to the position of output pixel, for large receptive field generated by 7×7 convolution to build a discriminative receptive field. The output combines the two receptive fields to build a two-layer AGD by directly assigning impacts of different scales on the receptive field. Sequentially, each LO expands and amplifies the receptive field of the previous LO by Amp and provides a discriminative receptive field for adding impacts of small-scale new pixels. The final receptive field of three-layer RFA results in a four-layer receptive field following an AGD. The ERF can be expanded and the AGD of ERF can be maintained through the stack of many RFA modules, as illustrated in Fig. 1 (D).

2.5. UniConvNet Model

The Three-layer RFA can be integrated into any model as a plug-and-play module, effectively replacing each convolution in ConvNet architectures. For better performance, we directly integrate the Three-layer RFA module into the state-of-the-art CNN-based model, InternImage [69]. Specifically, the convolution used in the 3×3 convolution residual component is the DCNV3 in the InternImage [69] and we remove the softmax normalization in DCNV3 as DCNV4 did because the optimizations for memory on DCNV4 in the Efficient Deformable ConvNets [72] is in-

Model	Type	Scale	#Params	FLOPs	Acc(%)	Publication
EdgeViT-XSS [47]	T	256 ²	4.1M	0.577G	74.4	ECCV'22
FastViT-T8 [64]	T	256 ²	3.6M	0.7G	75.6	ICCV'23
PVTv2-B0 [68]	T	224 ²	3.7M	0.572G	70.5	CVM'22
tiny-MOAT-0 [74]	T	224 ²	3.4M	0.8G	75.5	ICLR'23
UniRepLKNet-A [18]	C	224 ²	4.4M	0.6G	77.0	CVPR'24
StarNet-S2 [43]	C	224 ²	3.7M	0.547G	74.8	CVPR'24
UniConvNet-A(ours)	C	224²	3.4M	0.589G	77.0	-
EdgeNeXT-S [44]	T	256 ²	5.6M	0.965G	78.8	ECCV'22
MobileViTv1-S [45]	T	256 ²	5.6M	2.01G	78.4	ICLR'22
tiny-MOAT-1 [74]	T	224 ²	5.1M	1.2G	78.3	ICLR'23
EMO-5M [81]	T	224 ²	5.1M	0.903G	78.4	ICCV'23
StarNet-S3 [43]	C	224 ²	5.8M	0.757G	77.3	CVPR'24
RepViT-M0.9 [66]	C	224 ²	5.1M	0.8G	77.4	CVPR'24
DCNV4 [72]	C	224 ²	5.3M	0.805G	78.5	CVPR'24
UniConvNet-P0(ours)	C	224²	5.2M	0.832G	79.1	-
EdgeViT-XS [47]	T	256 ²	6.7M	1.136G	77.5	ECCV'22
FastViT-T12 [64]	T	256 ²	6.8M	1.4G	79.1	ICCV'23
SMT-M [36]	T	224 ²	6.5M	1.3G	78.4	ICCV'23
EMO-6M [81]	T	224 ²	6.1M	0.961G	79.0	ICCV'23
StarNet-S4 [43]	C	224 ²	7.5M	1.075G	78.4	CVPR'24
RepViT-M1.0 [66]	C	224 ²	6.8M	1.1G	78.6	CVPR'24
UniRepLKNet-F [18]	C	224 ²	6.2M	0.9G	78.6	CVPR'24
UniConvNet-P1(ours)	C	224²	6.1M	0.895G	79.6	-
FastViT-S12 [64]	T	256 ²	8.8M	1.8G	79.8	ICCV'23
EfficientNet-B2 [59]	C	288 ²	9.2M	1.0G	80.1	ICCV'19
RepViT-M1.1 [66]	C	224 ²	8.2M	1.3G	79.4	CVPR'24
UniConvNet-P2(ours)	C	224²	7.6M	1.25G	80.5	-
Shunted-T [52]	T	256 ²	11.5M	2.1G	79.8	CVPR'22
EdgeViT-S [47]	T	256 ²	11.1M	1.9G	81.0	ECCV'22
FastViT-SA12 [64]	T	256 ²	10.9M	1.9G	80.6	ICCV'23
EfficientFormer-L1 [33]	T	224 ²	12.3M	1.3G	79.1	NeurIPS'22
MpViT-XS [31]	T	224 ²	10.5M	2.9G	80.9	CVPR'22
EfficientNet-B3 [59]	C	300 ²	12.0M	1.8G	81.6	ICCV'19
UniConvNet-N0(ours)	C	224²	10.2M	1.65G	81.6	-
PVTv1-Tiny [67]	T	224 ²	13.2M	1.9G	75.1	ICCV'21
RepViT-M1.5 [66]	C	224 ²	14.0M	2.3G	81.2	CVPR'24
UniConvNet-N1(ours)	C	224²	13.1M	1.88G	82.2	-
FastViT-SA24 [64]	T	256 ²	20.6M	3.8G	82.6	ICCV'23
Dilate-T [29]	T	224 ²	17.0M	3.2G	82.1	TMM'23
RepViT-M2.3 [66]	C	224 ²	22.9M	4.5G	82.5	CVPR'24
UniRepLKNet-N [18]	C	224 ²	18.3M	2.8G	81.6	CVPR'24
UniConvNet-N2(ours)	C	224²	15.0M	2.47G	82.7	-
XCiT-S12/16 [1]	T	224 ²	26.0M	4.8G	82.0	NeurIPS'21
ViTAE-S [73]	T	224 ²	23.6M	5.6G	82.0	NeurIPS'21
CoAtNet-0 [13]	T	224 ²	25.0M	4.0G	81.6	NeurIPS'21
PVTv2-B2-Li [68]	T	224 ²	22.6M	3.9G	82.1	CVM'22
HorNet-T [50]	C	224 ²	23.0M	4.0G	83.0	NeurIPS'22
UniConvNet-N3(ours)	C	224²	19.7M	3.37G	83.2	-

Table 1. Image classification performance on the ImageNet validation set for lightweight variants. “Type” refers to model type, where “C” and “T” denote pure CNN and the model using the Transformer for its architecture, respectively. “Scale” is the input image scale. “Acc” is the TOP-1 accuracy.

compatible with the other CNN-based models. We develop a new CNN-based backbone model called UniConvNet using various stacking strategies for its variants. The overall architecture and stacking rules are illustrated in Fig. 5. We present integral blocks of the model and provide details on the embedded blocks in Appendix C.

3. Experiments

We develop different UniConvNet variants to match the complexities of various contemporary models, including state-of-the-art lightweight networks [43, 64, 66, 81] and large-scale networks [18, 38, 40, 69, 72, 74]. We evaluate the performance of UniConvNet variants and compare them to leading CNNs and ViTs across representative vision tasks, including image classification, object detection, and instance and semantic segmentation.

Model	Type	Scale	#Params	FLOPs	Acc(%)	Publication
CoAtNet-1 [13]	T	224 ²	42.0M	8.0G	83.3	NeurIPS'21
EfficientFormer-L3 [33]	T	224 ²	31.3M	3.9G	82.4	NeurIPS'22
Swin-T [38]	T	224 ²	29.0M	5.0G	81.3	ICCV'21
Focal-T [75]	T	224 ²	29.0M	5.0G	82.2	NeurIPS'21
CrossViT-18 [4]	T	224 ²	28.2M	6.1G	82.3	ICCV'21
UniRepLKNet-T [18]	C	224 ²	31.0M	4.9G	83.2	CVPR'24
SLaK-T [37]	C	224 ²	30.0M	5.0G	82.5	ICLR'23
InternImage-T [69]	C	224 ²	30.0M	5.0G	83.5	CVPR'23
FlashInternImage-T [72]	C	224 ²	30.0M	-	83.6	CVPR'24
PeLk-T [5]	C	224 ²	29.0M	5.6G	82.6	CVPR'24
ConvNeXt-T [40]	C	224 ²	29.0M	5.0G	82.1	CVPR'22
MogaNet-S [32]	C	224 ²	25.0M	5.0G	83.4	ICLR'24
WTConvNeXt-T [20]	-	224 ²	30.0M	4.5G	82.5	ECCV'24
UniConvNet-T(ours)	C	224²	30.3M	5.1G	84.2	-
SwinV2-S/8 [39]	T	256 ²	50.0M	12.0G	83.7	CVPR'22
CoAtNet-2 [13]	T	224 ²	75.0M	16.0G	84.1	NeurIPS'21
PVTv2-B4 [68]	T	224 ²	63.0M	10.0G	83.6	CVM'22
Swin-S [38]	T	224 ²	50.0M	9.0G	83.0	ICCV'21
RepLKNet-31B [17]	C	224 ²	79.0M	15.0G	83.5	CVPR'22
PeLk-S [5]	C	224 ²	50.0M	10.7G	83.9	CVPR'24
SLaK-S [37]	C	224 ²	55.0M	10.0G	83.8	ICLR'23
ConvNeXt-S [40]	C	224 ²	50.0M	9.0G	83.1	CVPR'22
HorNet-S [50]	C	224 ²	50.0M	9.0G	84.0	NeurIPS'22
InternImage-S [69]	C	224 ²	50.0M	8.0G	84.2	CVPR'23
FlashInternImage-S [72]	C	224 ²	50.0M	-	84.4	CVPR'24
WTConvNeXt-S [20]	-	224 ²	54.0M	8.8G	83.6	ECCV'24
UniConvNet-S(ours)	C	224²	50.0M	8.48G	84.5	-
SwinV2-B/8 [39]	T	256 ²	88.0M	20.0G	84.2	CVPR'22
Swin-B [38]	T	224 ²	88.0M	15.0G	83.5	ICCV'21
PVTv2-B5 [68]	T	224 ²	82.0M	12.0G	83.8	CVM'22
ConvNeXt-B [40]	C	224 ²	88.0M	15.0G	83.8	CVPR'22
InternImage-B [69]	C	224 ²	97.0M	16.0G	84.9	CVPR'23
FlashInternImage-B [72]	C	224 ²	97.0M	-	84.9	CVPR'24
SLaK-B [37]	C	224 ²	95.0M	17.0G	84.0	ICLR'23
PeLk-B [5]	C	224 ²	89.0M	18.3G	84.2	CVPR'24
HorNet-B [50]	C	224 ²	88.0M	16.0G	84.3	NeurIPS'22
WTConvNeXt-B [20]	-	224 ²	93.0M	15.5G	84.1	ECCV'24
UniConvNet-B(ours)	C	224²	97.6M	15.9G	85.0	-
Swin-B [38]	T	384 ²	88.0M	47.1G	84.5	ICCV'21
ConvNeXt-B [40]	C	384 ²	89.0M	45.0G	85.1	CVPR'22
UniConvNet-T(ours)	C	384²	30.3M	15.0G	85.4	-
ConvNeXt-L [40]	C	384 ²	198.0M	101.0G	85.5	CVPR'22
PeLk-L [5]	C	384 ²	89.0M	54.0G	85.6	CVPR'24
UniConvNet-S(ours)	C	384²	50.0M	24.9G	85.7	-
PeLk-B-101 [5]	C	384 ²	90.0M	68.3G	85.8	CVPR'24
UniConvNet-B(ours)	C	384²	97.6M	46.6G	85.9	-
SwinV2-L/24 [39]	T	384 ²	197.0M	115.0G	87.6	CVPR'22
Swin-L [38]	T	384 ²	197.0M	104.0G	87.3	ICCV'21
MOAT-3 [74]	T	384 ²	190.0M	141.2G	88.2	ICLR'23
HorNet-L [50]	C	384 ²	202.0M	102.0G	87.7	NeurIPS'22
ConvNeXt-L [40]	C	384 ²	198.0M	101.0G	87.5	CVPR'22
MogaNet-XL [32]	C	384 ²	181.0M	102.0G	87.8	ICLR'24
RepLKNet-31L [17]	C	384 ²	172.0M	96.0G	86.6	CVPR'22
UniConvNet-L(ours)	C	384²	201.8M	100.1G	88.2	-
CoAtNet-4 [13]	T	384 ²	275.0M	190.0G	87.9	NeurIPS'21
ConvNeXt-XL [40]	C	384 ²	350.0M	179.0G	87.7	CVPR'22
InternImage-XL [69]	C	384 ²	335.0M	163.0G	88.0	CVPR'23
FlashInternImage-L [72]	C	384 ²	224.0M	-	88.1	CVPR'24
InternImage-L [69]	C	384 ²	223.0M	108.0G	87.7	CVPR'23
UniConvNet-XL(ours)	C	384²	226.7M	115.2G	88.4	-

Table 2. **Image classification performance on the ImageNet validation set for scaled-up variants.** “Type” refers to model type, where “C” and “T” denote pure CNN and the model using the Transformer for its architecture, respectively. “Scale” is the input image scale. “Acc” is the TOP-1 accuracy. “†” indicates the model is pre-trained on ImageNet-22K[14].

3.1. Image Classification

For a fair comparison, in line with common practices [40, 72], UniConvNet-A/P0/P1/P2/N0/N1/N2/N3/T/S/B are trained on ImageNet-1K for 300 epochs, while UniConvNet-L is first trained on ImageNet-22K for 90 epochs and then fine-tuned on ImageNet-1K for 20 epochs. Detailed ImageNet-1K/22K training settings, ImageNet-1K fine-tune settings and training recipes for different variants are presented in Appendix D.1, D.2 and D.3, respectively.

heavy RetinaNet									
Backbone	Type	#Params	FLOPs	mAP	mAP ₅₀	mAP ₇₅	mAP _S	mAP _M	mAP _L
EdgeViT-XXS [47]	T	13.1M	-	38.7	59.0	41.0	22.4	42.0	51.6
EMO-5M [81]	T	14.4M	-	38.9	59.8	41.0	23.8	42.2	51.7
UniConvNet-A(ours)	C	12.6M	16.3G	40.0	60.4	42.9	23.3	44.2	53.2
MobileFormer-151M [8]	T	14.4M	-	34.2	53.4	36.0	19.9	36.8	45.3
MobileFormer-214M [8]	T	15.2M	-	35.8	55.4	38.0	21.8	38.5	46.8
MobileFormer-294M [8]	T	16.1M	-	36.6	56.6	38.6	21.9	39.5	47.9
EdgeViT-XS [47]	T	16.3M	-	40.6	61.3	43.3	25.2	43.9	54.6
UniConvNet-P0(ours)	C	14.4M	16.8G	41.1	61.4	43.9	24.2	45.2	55.2
MobileFormer-508M [8]	T	17.9M	-	38.0	58.3	40.3	22.9	41.2	49.7
UniConvNet-P2(ours)	C	16.9M	16.8G	42.2	62.5	45.2	25.1	45.9	56.2
PVTv1-Tiny [67]	T	23.0M	-	36.7	56.9	38.9	22.6	38.8	50.0
ResNet-18 [23]	C	21.3M	-	31.8	49.6	33.6	16.3	34.3	43.2
UniConvNet-N0(ours)	C	20.7M	18.8G	42.8	63.2	45.7	25.2	47.0	56.8
PVTv2-B1 [68]	T	23.8M	-	41.2	61.9	43.9	25.4	44.5	54.3
UniConvNet-N1(ours)	C	23.9M	19.4G	44.6	65.7	47.9	27.9	49.0	59.2
PVTv1-Small [67]	T	44.1M	-	40.4	61.3	43.0	25.0	42.9	55.7
MPViT-T [31]	T	28.0M	-	41.8	62.7	44.6	27.2	45.1	54.2
Twins-PCPVT-S [10]	T	34.4M	-	43.0	64.1	46.0	27.5	46.3	57.3
PVTv2-B2 [68]	T	35.1M	-	44.6	65.6	47.6	27.4	48.8	58.6
Shunted-S [52]	T	32.1M	-	45.4	65.9	49.2	28.7	49.3	60.0
UniConvNet-N2(ours)	C	26.0M	20.7G	45.5	66.4	48.9	28.4	50.2	60.4

light SSDLite									
Backbone	Type	#Params	FLOPs	mAP	mAP ₅₀	mAP ₇₅	mAP _S	mAP _M	mAP _L
MobileNetv3 [26]	C	5.0M	0.6G	22.0	-	-	-	-	-
MobileNetv2 [53]	C	4.3M	0.8G	22.1	-	-	-	-	-
MobileNetv1 [27]	C	5.1M	1.3G	22.2	-	-	-	-	-
MixNet [60]	T	4.5M	-	22.3	-	-	-	-	-
MNASNet [61]	T	4.9M	0.8G	23.0	-	-	3.8	21.7	42.0
MobileViT-Small [45]	T	5.7M	3.4G	27.7	-	-	-	-	-
EdgeNeXt-S [44]	T	6.2M	2.1G	27.9	-	-	-	-	-
UniConvNet-A(ours)	C	4.4M	1.3G	29.5	46.7	30.2	5.3	31.8	53.6
MobileViTv2-1.25 [46]	T	8.2M	4.7G	27.8	-	-	-	-	-
EMO-5M [81]	T	6.0M	1.8G	27.8	45.2	28.2	5.2	30.5	50.0
UniConvNet-P0(ours)	C	6.4M	1.8G	30.5	48.1	31.1	6.2	33.6	55.3
UniConvNet-P2(ours)	C	8.6M	2.6G	32.0	49.8	33.0	7.6	35.8	56.0
MobileViTv2-1.75 [46]	T	14.9M	9.0G	29.5	-	-	-	-	-
UniConvNet-N0(ours)	C	12.2M	3.8G	33.7	52.1	34.8	9.4	37.6	58.5

Table 3. **Object detection performance by heavy RetinaNet and light SSDLite on COCO val2017.** “Type” refers to model type, where “C” and “T” denote pure CNN and the model using the Transformer for its architecture, respectively. The FLOPs are measured with 320×320 inputs.

As is shown in Tab. 1 and Tab. 2, our proposed model variants demonstrate significant improvements over state-of-the-art models, effectively bridging the gap between lightweight and large-scale models. Existing models either exhibit inferior performance in lightweight scenarios or fail to achieve adequate accuracy when scaled up.

3.2. Downstream Tasks

Object Detection and Instance Segmentation on COCO We fine-tune the heavy RetinaNet [35] and light SSDLite [26] using our ImageNet-1K pre-trained UniConvNet on the MS-COCO 2017 with a $1 \times$ schedule training recipe. We further fine-tune scaled-up UniConvNet using the representative object detection framework Mask R-CNN [24] and Cascade Mask R-CNN [3] on the MS-COCO 2017 datasets, employing either a $1 \times$ (12epoch) training schedule or a $3 \times$ (36 epoch) training schedule. Detailed fine-tuning settings are provided in Appendix D.4.

Semantic Segmentation on ADE20K We fine-tune DeepLabv3 [7] and PSPNet [83] using our ImageNet-1K pre-trained UniConvNet on the ADE20K dataset with a training recipe of 160k iterations. We also fine-tune the scaled-up UniConvNet using the representative semantic segmentation framework UperNet on the ADE20K dataset for 160k iterations. Detailed fine-tuning settings are provided in Appendix D.5.

Backbone	Type	#Params	FLOPs	Mask R-CNN 1× schedule						Mask R-CNN 3× + MS schedule					
				AP^b	AP_{50}^b	AP_{75}^b	AP^m	AP_{50}^m	AP_{75}^m	AP^b	AP_{50}^b	AP_{75}^b	AP^m	AP_{50}^m	AP_{75}^m
UniConvNet-N2	C	34.7M	220G	46.6	68.0	51.3	41.9	65.1	45.2	48.4	69.7	53.2	43.2	66.7	46.4
PVTv2-B2 [68]	T	45.0M	309G	45.3	67.1	49.6	41.2	64.2	44.4	47.8	69.7	52.6	43.1	66.8	46.7
ViT-Adapter-S [9]	T	48.0M	403G	44.7	65.8	48.3	39.9	62.5	42.8	48.2	69.7	52.5	42.8	66.4	45.9
MogaNet-S [32]	C	45.0M	272G	46.7	-	-	42.2	-	-	-	-	-	-	-	-
UniConvNet-N3	C	39.4M	239G	47.0	68.6	51.8	42.4	65.6	45.7	49.4	70.7	54.4	44.2	67.9	47.5
Swin-T [38]	T	48.0M	267G	42.7	65.2	46.8	39.3	62.2	42.2	46.0	68.1	50.3	41.6	65.1	44.9
ConvNeXt-T [40]	C	48.0M	262G	44.2	66.6	48.3	40.1	63.3	42.8	46.2	67.9	50.8	41.7	65.0	44.9
InternImage-T [69]	C	49.0M	270G	47.2	69.0	52.1	42.5	66.1	45.8	49.1	70.4	54.1	43.7	67.3	47.3
MogaNet-B [32]	C	63.0M	373G	47.9	-	-	43.2	-	-	-	-	-	-	-	-
FlashInternImage-T [72]	C	49.0M	-	48.0	-	-	43.1	-	-	49.5	-	-	44.0	-	-
UniConvNet-T	C	50.0M	265G	48.2	69.8	52.9	43.3	66.6	45.7	50.1	71.0	54.8	44.5	68.4	48.0
Swin-S [38]	T	69.0M	354G	44.8	66.6	48.9	40.9	63.4	44.2	48.2	69.8	52.8	43.2	67.0	46.1
ConvNeXt-S [40]	C	70.0M	348G	45.4	67.9	50.0	41.8	65.2	45.1	47.9	70.0	52.7	42.9	66.9	46.2
InternImage-S [69]	C	69.0M	340G	47.8	69.8	52.8	43.3	67.1	46.7	49.7	71.1	54.5	44.5	68.5	47.8
FlashInternImage-S [72]	C	69.0M	-	49.2	-	-	44.0	-	-	50.5	-	-	44.9	-	-
UniConvNet-S	C	70.0M	336G	48.8	70.4	53.4	43.8	67.4	47.3	50.8	71.6	55.6	45.2	69.3	48.9
Swin-B [38]	T	107.0M	496G	46.9	-	-	42.3	-	-	48.6	70.0	53.4	43.3	67.1	46.7
PVTv2-B5 [68]	T	102.0M	557G	47.4	68.6	51.9	42.5	65.7	46.0	48.4	69.2	52.9	42.9	66.6	46.2
ConvNeXt-B [40]	C	108.0M	486G	47.0	69.4	51.7	42.7	66.3	46.0	48.5	70.1	53.3	43.5	67.1	46.7
InternImage-B [69]	C	115.0M	501G	48.8	70.9	54.0	44.0	67.8	47.4	50.3	71.4	55.3	44.8	68.7	48.0
FlashInternImage-B [72]	C	115.0M	-	50.1	-	-	44.5	-	-	50.6	-	-	45.4	-	-
UniConvNet-B	C	118.0M	498G	50.0	71.7	55.3	45.0	69.0	48.5	51.2	72.2	56.1	45.6	69.6	49.2

Backbone	Type	#Params	FLOPs	Cascade Mask R-CNN 1× schedule						Cascade Mask R-CNN 3× + MS schedule					
				AP^b	AP_{50}^b	AP_{75}^b	AP^m	AP_{50}^m	AP_{75}^m	AP^b	AP_{50}^b	AP_{75}^b	AP^m	AP_{50}^m	AP_{75}^m
Swin-L [38]	T	253.0M	1382G	51.8	71.0	56.2	44.9	68.4	48.9	53.9	72.4	58.8	46.7	70.1	50.8
RepLkNet-31L [17]	C	229.0M	1321G	-	-	-	-	-	-	53.9	72.5	58.6	46.5	70.0	50.6
ConvNeXt-L [40]	C	255.0M	1354G	53.5	72.8	58.3	46.4	70.2	50.2	54.8	73.8	59.8	47.6	71.3	51.7
ConvNeXt-XL [40]	C	407.0M	1898G	53.6	72.9	58.5	46.5	70.3	50.5	55.2	74.2	59.9	47.7	71.6	52.2
HorNet-L [50]	C	259.0M	1358G	-	-	-	-	-	-	56.0	-	-	48.6	-	-
InternImage-L [69]	C	277.0M	1399G	54.9	74.0	59.8	47.7	71.4	52.1	56.1	74.8	60.7	48.5	72.4	53.0
InternImage-XL [69]	C	387.0M	1782G	55.3	74.4	60.1	48.1	71.9	52.4	56.2	75.0	61.2	48.8	72.5	53.4
FlashInternImage-L [72]	C	277.0M	-	55.6	-	-	48.2	-	-	56.7	-	-	48.9	-	-
UniConvNet-L [†]	C	254.8M	1288G	55.7	74.4	60.4	48.3	71.9	52.9	56.6	75.6	61.8	48.9	73.0	53.4

Table 4. Object detection and instance segmentation performance by Mask R-CNN and Cascade Mask R-CNN on COCO val2017. “Type” refers to model type, where “C” and “T” denote pure CNN and the model using the Transformer for its architecture, respectively. The FLOPs are measured with 1280×800 inputs.

DeepLabv3				
Model	Type	#Params	FLOPs	mIoU
MobileViTv2-0.5 [46]	T	6.3M	26.1G	31.9
EMO-2M [81]	T	6.9M	3.5G	35.3
UniConvNet-A(ours)	C	7.9M	4.2G	38.2
MobileViTv2-0.75 [46]	T	9.6M	40.0G	34.7
EMO-5M [81]	T	10.3M	5.8G	37.8
UniConvNet-P0(ours)	C	9.7M	5.3G	39.7
MobileViTv2-1.0 [46]	T	13.4M	56.4G	37.0
UniConvNet-P2(ours)	C	12.6M	7.8G	40.0
MobileNetv2 [53]	C	18.7M	75.4G	34.1
UniConvNet-N0(ours)	C	17.2M	11.0G	41.0
UniConvNet-N1(ours)	C	20.3M	12.3G	42.1
MobileViTv2-2.0 [46]	T	64.0M	147.0G	40.9
ResNet-50 [23]	C	68.2M	270.3G	42.4
UniConvNet-N2(ours)	C	22.5M	15.7G	42.9

PSPNet				
Model	Type	#Params	FLOPs	mIoU
EMO-2M [81]	T	5.5M	3.1G	34.5
MobileViTv2-0.75 [46]	T	6.2M	26.6G	35.2
UniConvNet-A(ours)	C	6.5M	3.8G	37.9
MobileViTv2-1.0 [46]	T	9.4M	40.3G	36.5
EMO-5M [81]	T	8.5M	5.3G	38.2
UniConvNet-P0(ours)	C	8.1M	4.8G	39.2
MobileNetv2 [53]	T	13.7M	53.1G	29.7
UniConvNet-P2(ours)	C	10.9M	7.3G	39.6
UniConvNet-N0(ours)	C	15.0M	10.4G	40.1
MobileViTv2-1.75 [46]	T	22.5M	95.9G	39.8
UniConvNet-N1(ours)	C	18.2M	11.6G	42.1
ResNet-50 [23]	T	49.1M	179.1G	41.1
UniConvNet-N2(ours)	C	20.3M	15.1G	42.5

Table 5. Semantic segmentation performance by DeepLabv3 and PSPNet on ADE20K dataset. “Type” refers to model type, where “C” and “T” denote pure CNN and the model using the Transformer for its architecture, respectively. The FLOPs are measured with 512×512 inputs.

Backbone	Type	Crop Size	#Params	FLOPs	mIoU(SS)	mIoU(MS)
Swin-T [38]	T	512^2	60.0	939	44.5	45.8
ConvNeXt-T [40]	C	512^2	60.0	939	46.0	46.7
UniConvNet-N2	C	512^2	43.9	893	47.9	48.9
SLaK-T [37]	C	512^2	65.0	936	47.6	-
InternImage-T [69]	C	512^2	59.0	944	47.9	48.1
UniConvNet-N3	C	512^2	48.6	912	49.0	50.0
Shunted-S [52]	T	512^2	52.0	940	48.9	49.9
PeLk-T [5]	C	512^2	62.0	970	48.1	-
UniRepLkNet-T [18]	C	512^2	61.0	946	48.6	49.1
FlashInternImage-T [72]	C	512^2	59.0	944	49.3	50.3
MogaNet-B [32]	C	512^2	74.0	1050	50.1	-
UniConvNet-T	C	512^2	59.2	939	50.3	51.2
Swin-S [38]	T	512^2	81.0	1038	47.6	49.5
ConvNeXt-S [40]	C	512^2	82.0	7027	48.7	49.6
SLaK-S [37]	C	512^2	91.0	1028	49.4	-
PeLk-S [5]	C	512^2	84.0	1077	49.7	-
InternImage-S [69]	C	512^2	80.0	1017	50.1	50.9
UniRepLkNet-S [18]	C	512^2	86.0	1036	50.5	51.0
FlashInternImage-S [69]	C	512^2	80.0	-	50.6	51.6
MogaNet-L [32]	C	512^2	113.0	1176	50.9	-
UniConvNet-S	C	512^2	78.9	1015	52.2	52.8
Swin-B [38]	T	512^2	121.0	1188	48.1	49.7
ConvNeXt-B [40]	C	512^2	122.0	1170	49.1	49.9
RepLkNet-31B [17]	C	512^2	112.0	1170	49.9	50.6
SLaK-B [37]	C	512^2	135.0	1172	50.2	-
PeLk-B-101 [5]	C	512^2	126.0	1339	50.6	-
InternImage-B [69]	C	512^2	128.0	1185	50.8	51.3
FlashInternImage-B [69]	C	512^2	128.0	-	52.0	52.6
UniConvNet-B	C	512^2	126.5	1179	52.3	52.9
Swin-L [38]	T	640^2	234	2468	52.1	53.5
RepLkNet-31L [17]	C	640^2	207	2404	52.4	52.7
ConvNeXt-L [40]	C	640^2	235	2458	53.2	53.7
ConvNeXt-XL [40]	C	640^2	391	3335	53.6	54.0
InternImage-L [69]	C	640^2	256	2526	53.9	54.1
InternImage-XL [69]	C	640^2	368	3142	55.0	55.3
UniConvNet-L [†]	C	640^2	234	2310	55.1	55.4

Table 6. Semantic segmentation performance by UperNet on ADE20K validation set. “Type” refers to model type, where “C” and “T” denote pure CNN and the model using the Transformer for its architecture, respectively. “SS” and “MS” denote single-scale and multi-scale testing, respectively. The FLOPs are measured with 512×2048 or 640×2560 inputs.

Overall Results As shown in Tab. 3, Tab. 4, Tab. 5 and Tab. 6, our proposed UniConvNet significantly enhances

Model	N	Kernel Size	#Params	FLOPs	Acc(%)
UniConvNet-A	3	5, 7, 9	3.5M	0.564G	76.6
	3	7, 9, 11	3.4M	0.589G	77.0
	3	9, 11, 13	3.5M	0.579G	76.9
UniConvNet-P0	3	5, 7, 9	5.1M	0.810G	78.8
	3	7, 9, 11	5.2M	0.832G	79.1
	3	9, 11, 13	5.3M	0.868G	79.3
UniConvNet-T	3	11, 13, 15	5.1M	0.845G	78.8
	3	5, 7, 9	30.0M	5.0G	84.1
	3	7, 9, 11	30.3M	5.1G	84.2
UniConvNet-N0	3	9, 11, 13	29.6M	5.0G	84.1
	3	7, 9, 11	10.2M	1.65G	81.6
	4	5, 7, 9, 11	9.8M	1.64G	81.5
UniConvNet-N2	4	7, 9, 11, 13	10.0M	1.70G	81.3
	3	27, 29, 31	14.9M	2.87G	81.8
	3	7, 9, 11	15.0M	2.47G	82.7
UniConvNet-A	3	7, 11, 31	3.56M	0.687G	76.0
	3	7, 9, 29	3.45M	0.647G	75.9
	3	7, 9, 11	3.4M	0.589G	77.0
UniConvNet-P1	3	7, 11, 31	6.2M	1.162G	79.4
	3	7, 9, 11	6.1M	0.895G	79.6

Table 7. **Ablation studies on layer number and progressive kernel size.** “Acc” is the TOP-1 accuracy. “N” represents the number in the pyramid order.

performance compared to state-of-the-art models, offering lighter parameters and reduced FLOPs. This demonstrates the effectiveness and efficiency of the proposed Three-layer RFA and its improved capability for downstream tasks.

4. Analysis

4.1. Three-layer RFA properly expand ERF while maintaining AGD

Ablation Studies on Layer Number and Kernel Size Section 2.3 presents the constraints on layer N and progressive kernel size. Based on UniConvNet-T, we adjust layer N and progressive kernel size for ablation studies. The results indicate that the proposed principles for these two hyperparameters are effective and accurate. As illustrated in Tab. 7, for layer 3, progressive kernel sizes smaller than (5, 7, 9) may be insufficient for expanding ERF to a level of existing large-kernel ConvNets and result in inferior performance. Progressive kernel sizes larger than (9, 11, 13) achieve better TOP-1 accuracy than UniConvNet-P0, with slightly higher parameters and FLOPs. In contrast, UniConvNet-A and UniConvNet-T perform better with kernel sizes of (7, 9, 11). Progressive kernel sizes of (9, 11, 13) are inefficient for constructing deep models with equivalent parameters, which is crucial for model perception. Therefore, we choose a kernel size of (7, 9, 11) for efficiency. For layer 4, progressive kernels result in a theoretical receptive field much larger than the image size of 14×14 in stage 3, which is wasteful for an image resolution of 224×224 . It contradicts the original goal of alleviating parameters and FLOPs burden in contemporary large-kernel ConvNets.

Investigation on different AGD of ERF We also examine the perceptual capabilities of large kernel sizes such as

UniConvNet	Overall	RFA ³⁻¹	Modified DCONV3	Feed-Forward	Class Head
UniConvNet-A	3.4/0.589	0.747/0.157	0.91/0.161	1.331/0.234	0.409/0.037
UniConvNet-P0	5.17/0.832	1.06/0.205	1.41/0.229	2.06/0.334	0.64/0.066
UniConvNet-P1	6.1/0.895	1.26/0.22	1.70/0.247	2.495/0.362	0.635/0.064
UniConvNet-P2	7.57/1.25	1.625/0.315	2.16/0.354	3.153/0.516	0.648/0.065
UniConvNet-N0	10.23/1.65	2.18/0.405	2.16/0.354	4.62/0.851	1.27/0.04
UniConvNet-N1	13.06/1.88	2.85/0.458	2.85/0.409	6.095/0.867	1.275/0.134
UniConvNet-N2	15.0/2.47	3.34/0.622	3.31/0.546	7.08/1.161	1.27/0.139
UniConvNet-N3	19.7/3.37	4.54/0.849	4.45/0.762	9.48/1.619	1.25/0.142
UniConvNet-T	30.3/5.1	5.41/1.06	5.74/1.0	14.61/2.46	4.54/0.58
UniConvNet-S	50/8.48	4.54/0.849	4.45/0.762	9.48/1.619	1.25/0.142
UniConvNet-B	97.6/15.9	10.89/1.97	11.44/1.95	24.61/4.18	3.06/0.38
UniConvNet-L	201.8/100.1	22.01/21.79	25.77/23.48	80.24/50.56	11.11/4.28
UniConvNet-XL	226.7/115.2	24.7/25.07	28.95/27.02	90.14/58.19	12.48/4.93

Table 8. **#Parameters and FLOPs distributions of UniConvNet variants.** #Parameters(M)/FLOPs(G) are #parameters and FLOPs for each block, respectively (e.g., 3.4/0.589 are the overall #parameters and FLOPs for UniConvNet-A, respectively).

(27, 29, 31) as used in RepLKNet [17]. The performance results show that using large kernels is neither efficient nor effective for constructing long-range ERF following AGD, disrupting AGD on small-scale pixels. We further use relatively small kernels in the first two layers, then use an extremely large kernel in the latter layer, such as (7, 9, 29) and (7, 11, 31). This establishes a small-scale AGD in a smaller area, compared to the kernel sizes of (7, 9, 11), and expands the ERF by the latter extremely large kernel. The inferior performance demonstrates that a large ERF with continuous AGD, generated by three-layer RFA, from center to edge is vital and proper. We analyze the ERF of several models and demonstrate that a proper AGD of small-scale pixels is more important than expanding the ERF. Please refer to Appendix A for detailed analysis.

4.2. Three-layer RFA is Efficient and Effective

Efficiency We present the parameters and FLOPs of UniConvNet variants in Tab. 8. UniConvNet comprises Three-layer RFA, modified DCONV3, a feed-forward layer, and a class head. Generally, compared to small-kernel modified DCONV3, Three-layer RFA has fewer or comparable parameters and FLOPs. This suggests that our proposed Three-layer RFA can establish long-range dependencies while reducing FLOP costs and enhancing parameter efficiency.

Effectiveness We perform an ablation study comparing various combinations of large-kernel and small-kernel convolutions to evaluate the effectiveness of different modules. As illustrated in Tab. 9, models using only Three-layer RFA or Modified DCONV3 achieve similar top-1 accuracies of 78.4 and 78.5, respectively. This indicates that the proposed Three-layer RFA retains comparable feature perception capabilities without relying on the basic small-scale information typically used in conventional ConvNets. Additionally, we replace Modified DCONV3 in UniConvNet-P0 with a depth-wise 3×3 convolution. The top-1 accu-

Model	#Params	Large Kernel	Small Kernel	Acc(%)
UniConvNet-P0	5.2M	Three-layer RFA	Modified DCNV3	79.1
FlashInternImage [72]	5.2M	Three-layer RFA	×	78.4
	5.3M	×	DCNV4	78.5
ConvNeXt [40]	5.1M	Three-layer RFA	DW 3×3	78.9
	5.2M	DW 7×7	DW 3×3	77.0
	5.3M	×	DW 3×3	77.0
UniConvNet-N2	15.0M	Three-layer RFA	Modified DCNV3	82.7
FlashInternImage [72]	15.3M	×	DCNV4	82.2
	14.9M	Three-layer RFA	DW 3×3	81.9
ConvNeXt [40]	15.1M	DW 7×7	DW 3×3	81.0
	30.3M	Three-layer RFA	Modified DCNV3	84.2
UniConvNet-T	30.3M	×	DCNV4	83.6
FlashInternImage-T [72]	30.0M	×	DCNV4	83.6
	30.7M	Three-layer RFA	DW 3×3	83.7
ConvNeXt-T [40]	29.0M	DW 7×7	DW 3×3	82.1

Table 9. Ablation comparisons of different large-kernel and small-kernel convolutions. “Acc” is the TOP-1 accuracy.

Model	#Params	Scale	Acc(%)	Throughput
Swin-T [38]	29.0M	224 ²	81.3	1989/3619
ConvNeXt-T [40]	29.0M	224 ²	82.1	2485/4305
InternImage-T [69]	30.0M	224 ²	83.5	1409/1746
UniConvNet-T	30.3M	224 ²	84.2	1480/1825
ConvNeXt-XL [40]	350.0M	224 ²	87.7	170/299
InternImage-XL [69]	335.0M	224 ²	88.0	125/174
InternImage-L [69]	223.0M	224 ²	87.7	158/214
UniConvNet-XL	226.7M	224 ²	88.4	168/228

Table 10. Image classification throughput on ImageNet-1K. “Acc(%)” is the TOP-1 Accuracy. The overall throughput of each model, measured as the number of images processed per second, is reported in FP32/FP16 data formats.

racy is 78.9, 0.2 lower than UniConvNet-P0, indicating that UniConvNet-P0 is slightly improved by Modified DCNV3 compared to when it uses simple depth-wise 3×3 convolution. In contrast, replacing Three-layer RFA with another large-kernel convolution (depth-wise 7×7 convolution) results in a drop in top-1 accuracy to 77.0, highlighting the effectiveness of our proposed Three-layer RFA. Furthermore, models using only depth-wise 3×3 convolution achieve a top-1 accuracy of 77.0, indicating that depth-wise 7×7 convolution does not enhance the model’s perception capabilities. This further validates the effectiveness of the proposed Three-layer RFA. The ablation studies on models of 15M and 30M parameters manifest a consistent performance improvement ability of the proposed Three-layer RFA.

4.3. Throughput analysis

We benchmark throughput on several classic and relevant models. As shown in Tab. 10, UniConvNets improve the throughput compared to InternImage [69] with similar 3×3 convolution. Please refer to Appendix B for detailed configurations.

5. Related Work

Convolutional neural networks (ConvNets) [16, 21, 23, 26, 27, 43, 51, 53, 54, 69, 71, 72, 78, 80] have long been the standard architecture for vision recognition due to their intrinsic inductive bias. CNNs utilize a stack of small kernels to establish local dependencies, which limits their

ability to perceive. However, their dominance is being challenged by the emergence of attention mechanisms in computer vision. In recent years, attention-based models [1, 4, 8, 9, 19, 29, 31, 33, 38, 44, 52, 64, 67, 68, 76] have gradually become essential for computer vision tasks due to their ability to establish global perception by building long-range dependencies through self-attention.

Inspired by this, large kernels are increasingly recognized for their ability to establish long-range dependencies in CNNs, thereby improving the accuracy of various vision tasks. The use of large kernels in convolutional networks originated with AlexNet [30] and Inception [55, 56, 58], which utilized 7×7 or 11×11 kernels in the low-level layers. ConvNeXt [40] examines the feasibility of large-kernel convolution within conventional ResNet-like architectures, where performance saturates at a kernel size of 7×7 .

Recently, RepLKNet [17] first scales the convolution kernel up to 31×31 using structural re-parameterization techniques and provides several guidelines for architectural design. VAN [22] acquires a 21×21 receptive field to establish long-term dependencies through sequential stacks of large-kernel depth-wise convolutions (DWConv) and depth-wise dilation convolutions. SLak [37] constructs a pure CNN architecture using sparse factorized 51×51 kernels, consisting of parallel 51×5 and 5×51 kernels. MogaNet [32] employs a spatial aggregation block utilizing 5×5 and 7×7 convolutions to adaptively aggregate discriminative features. PeLK [5] employs a human-like peripheral convolution to reduce parameters via parameter sharing while scaling the kernel size to a substantial 101×101 . However, recent large-kernel ConvNets suffer from parameters and FLOPs burden and disrupted AGD of ERF. In this work, we focus on expanding ERF while maintaining AGD for ConvNets of Any Scale.

6. Conclusion

We introduce a Receptive Field Aggregator (RFA) to expand effective receptive field (ERF) while maintaining the Asymptotically Gaussian Distribution (AGD) of ERF. Accordingly, we design a Three-layer RFA for input images with a resolution of 224×224 , which can be a plug-and-play module for ConvNets or replace the convolutional layers within them. Based on these designs, we propose a **universal convolutional neural network** (ConvNet), termed UniConvNet, and evaluate its performance across a wide range of vision recognition tasks. All variants of UniConvNet demonstrate superior performance with reduced parameters and FLOPs. This work may draw attention to the design of large ERF following an AGD, enhancing the ConvNet of any scale.

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